

Time Series — Solutions

Practice Exam — Week 10: Forecasting
EPFL, MATH-342 · Prof. S. C. Olhede

Problem 1 (5 points)

(a) For the causal AR(1) the MA(∞) representation is obtained by inverting $1 - \phi B$:

$$X_t = \frac{1}{1 - \phi B} \varepsilon_t = \sum_{j=0}^{\infty} \phi^j \varepsilon_{t-j}, \quad \text{so} \quad \psi_j = \phi^j \quad (j \geq 0).$$

Because the process is Gaussian, the best linear predictor coincides with the conditional expectation $\mathbb{E}[X_{n+h} \mid X_1, \dots, X_n]$. Iterating the recursion $X_{n+h} = \phi X_{n+h-1} + \varepsilon_{n+h}$ and taking conditional expectations (every future innovation $\varepsilon_{n+1}, \dots, \varepsilon_{n+h}$ has conditional mean 0, being uncorrelated with the data),

$$\begin{aligned} X_{n+h}^n &= \mathbb{E}[\phi X_{n+h-1} + \varepsilon_{n+h} \mid X_1, \dots, X_n] = \phi \mathbb{E}[X_{n+h-1} \mid X_1, \dots, X_n] \\ &= \phi^2 \mathbb{E}[X_{n+h-2} \mid X_1, \dots, X_n] = \dots = \phi^h X_n. \end{aligned}$$

Since $|\phi| < 1$, $X_{n+h}^n = \phi^h X_n \xrightarrow{h \rightarrow \infty} 0$, the process mean: the forecast *reverts to the mean*.

(b) By the ψ -weight theorem the forecast error is the sum of the unrealised future innovations,

$$e_n(h) = X_{n+h} - X_{n+h}^n = \sum_{j=0}^{h-1} \psi_j \varepsilon_{n+h-j} = \sum_{j=0}^{h-1} \phi^j \varepsilon_{n+h-j}.$$

The innovations are uncorrelated with common variance σ^2 , so

$$\text{Var}(e_n(h)) = \sigma^2 \sum_{j=0}^{h-1} \psi_j^2 = \sigma^2 \sum_{j=0}^{h-1} \phi^{2j} = \sigma^2 \frac{1 - (\phi^2)^h}{1 - \phi^2} = \sigma^2 \frac{1 - \phi^{2h}}{1 - \phi^2},$$

using the finite geometric sum $\sum_{j=0}^{h-1} r^j = (1 - r^h)/(1 - r)$ with $r = \phi^2$. As $h \rightarrow \infty$, $\phi^{2h} \rightarrow 0$ (since $|\phi| < 1$), so

$$\text{Var}(e_n(h)) \longrightarrow \frac{\sigma^2}{1 - \phi^2} = \gamma_0,$$

the marginal variance of the AR(1). The forecast-error variance increases monotonically in h and saturates at γ_0 : forecasting far ahead is no better than quoting the unconditional distribution.

(c) With $\phi = 0.6$ and $\sigma^2 = 4$:

$$X_{n+2}^n = \phi^2 X_n = 0.36 X_n.$$

The forecast-error variances are

$$\text{Var}(e_n(1)) = \sigma^2 \psi_0^2 = \sigma^2 = 4, \quad \text{Var}(e_n(2)) = \sigma^2(1 + \phi^2) = 4(1 + 0.36) = 4 \cdot 1.36 = 5.44.$$

(For reference $\gamma_0 = 4/(1 - 0.36) = 4/0.64 = 6.25$, and indeed $4 < 5.44 < 6.25$.) A 95% Gaussian prediction interval for X_{n+2} is

$$X_{n+2}^n \pm z_{0.975} \sqrt{\text{Var}(e_n(2))} = 0.36 X_n \pm 1.96 \sqrt{5.44} = 0.36 X_n \pm 1.96(2.3324) = 0.36 X_n \pm 4.571.$$

Problem 2 (5 points)

(a) Expand the prediction MSE as a quadratic in β , using $\text{Var}(X) = \Gamma_n$ and $\text{Cov}(X_{n+h}, X) = \gamma_{[h]}$ (the k -th entry of $\text{Cov}(X_{n+h}, X)$ is $\text{Cov}(X_{n+h}, X_k) = \gamma_{n+h-k}$, which is exactly the k -th entry of $\gamma_{[h]}$):

$$\begin{aligned} S(\beta) &= \mathbb{E}[(X_{n+h} - \beta^\top X)^2] = \text{Var}(X_{n+h} - \beta^\top X) \\ &= \gamma_0 - 2\beta^\top \text{Cov}(X_{n+h}, X) + \beta^\top \text{Var}(X)\beta = \gamma_0 - 2\beta^\top \gamma_{[h]} + \beta^\top \Gamma_n \beta. \end{aligned}$$

This is a quadratic in β bounded below by 0 (Γ_n is positive semi-definite), so its minimiser solves $\nabla_\beta S = 0$:

$$\nabla_\beta S = -2\gamma_{[h]} + 2\Gamma_n \beta = 0 \quad \implies \quad \boxed{\Gamma_n \beta = \gamma_{[h]}.}$$

Equivalence to orthogonality. Row k of $\Gamma_n \beta = \gamma_{[h]}$ reads

$$\sum_{j=1}^n \beta_j \gamma_{j-k} = \gamma_{n+h-k}, \quad k = 1, \dots, n.$$

On the other hand $\mathbb{E}[(X_{n+h} - \sum_j \beta_j X_j)X_k] = \text{Cov}(X_{n+h}, X_k) - \sum_j \beta_j \text{Cov}(X_j, X_k) = \gamma_{n+h-k} - \sum_j \beta_j \gamma_{j-k}$. Setting this to 0 for each k gives exactly the same n equations. Hence “error orthogonal to every observation” $\iff \Gamma_n \beta = \gamma_{[h]}$: the best linear predictor is the orthogonal projection of X_{n+h} onto $\text{span}\{X_1, \dots, X_n\}$.

(b) If Γ_n is invertible then $\beta = \Gamma_n^{-1} \gamma_{[h]}$, so

$$X_{n+h}^n = \beta^\top X = (\Gamma_n^{-1} \gamma_{[h]})^\top X = \gamma_{[h]}^\top \Gamma_n^{-1} X$$

(using $(\Gamma_n^{-1})^\top = \Gamma_n^{-1}$ since Γ_n is symmetric). For the MSE, substitute $\beta^\top \Gamma_n \beta = \beta^\top \gamma_{[h]}$ into $S(\beta)$:

$$P_{n+h}^n = \gamma_0 - 2\beta^\top \gamma_{[h]} + \beta^\top \Gamma_n \beta = \gamma_0 - 2\beta^\top \gamma_{[h]} + \beta^\top \gamma_{[h]} = \gamma_0 - \beta^\top \gamma_{[h]} = \gamma_0 - \gamma_{[h]}^\top \Gamma_n^{-1} \gamma_{[h]}.$$

(c) Based on X_1 (so $n = 1$, $h = 1$). The prediction equation $\Gamma_1 \beta = \gamma_{[1]}$ is the scalar equation $\gamma_0 \beta_1 = \gamma_1$, hence $\beta_1 = \gamma_1 / \gamma_0 = \rho_1$ and

$$X_2^1 = \beta_1 X_1 = \frac{\gamma_1}{\gamma_0} X_1 = \rho_1 X_1.$$

Based on X_1, X_2 (so $n = 2$, $h = 1$). Rather than invert Γ_2 , use the model: $X_3 = \phi_1 X_2 + \phi_2 X_1 + \varepsilon_3$, so the candidate predictor $\phi_1 X_2 + \phi_2 X_1$ has error

$$X_3 - (\phi_1 X_2 + \phi_2 X_1) = \varepsilon_3.$$

The future white noise ε_3 is uncorrelated with the past observations X_1, X_2 (which depend only on ε_s , $s \leq 2$), so

$$\mathbb{E}[(X_3 - \phi_1 X_2 - \phi_2 X_1)X_1] = \mathbb{E}[\varepsilon_3 X_1] = 0, \quad \mathbb{E}[(X_3 - \phi_1 X_2 - \phi_2 X_1)X_2] = \mathbb{E}[\varepsilon_3 X_2] = 0.$$

The error is orthogonal to both observations, so by the prediction-equations theorem of part (a) this candidate *is* the best linear predictor:

$$X_3^2 = \phi_1 X_2 + \phi_2 X_1.$$

(General fact: for an $\text{AR}(p)$ with $n \geq p$, the one-step forecast is simply $\sum_{i=1}^p \phi_i X_{n+1-i}$.)

Problem 3 (5 points)

(a) Consider any linear predictor $X_{n+h}^n = \sum_{i=1}^n \beta_i X_i$. Substituting the linear representation $X_i = \sum_{j=0}^{\infty} \psi_j \varepsilon_{i-j}$,

$$X_{n+h}^n = \sum_{i=1}^n \beta_i \sum_{j=0}^{\infty} \psi_j \varepsilon_{i-j} = \sum_{j=0}^{\infty} c_j \varepsilon_{n-j}$$

for some coefficients c_j : any predictor built from $X_1, \dots, X_n$ is a linear combination of the innovations up to time n (it “starts at ε_n ”). Now write $X_{n+h} = \sum_{j=0}^{\infty} \psi_j \varepsilon_{n+h-j}$ and split off the future innovations $\varepsilon_{n+1}, \dots, \varepsilon_{n+h}$ (indices $j = 0, \dots, h-1$). Since the ε ’s are uncorrelated with common variance σ^2 , the cross terms vanish and

$$\begin{aligned} \mathbb{E}[(X_{n+h} - X_{n+h}^n)^2] &= \mathbb{E}\left[\left(\sum_{j=0}^{\infty} \psi_j \varepsilon_{n+h-j} - \sum_{j=0}^{\infty} c_j \varepsilon_{n-j}\right)^2\right] \\ &= \sigma^2 \left(\sum_{j=0}^{h-1} \psi_j^2 + \sum_{j=h}^{\infty} (\psi_j - c_{j-h})^2 \right). \end{aligned}$$

The first sum collects the future innovations, which no past-based predictor can touch (it is *fixed*); the second sum matches ψ_j against c_{j-h} for the observable innovations. The second sum is minimised — to 0 — by choosing $c_{j-h} = \psi_j$, i.e. $c_j = \psi_{j+h}$. The predictor is then

$$X_{n+h}^n = \sum_{j=0}^{\infty} \psi_{j+h} \varepsilon_{n-j} = \sum_{j=h}^{\infty} \psi_j \varepsilon_{n+h-j},$$

the residual is $X_{n+h} - X_{n+h}^n = \sum_{j=0}^{h-1} \psi_j \varepsilon_{n+h-j}$, and the minimal MSE is

$$P_{n+h}^n = \sigma^2 \sum_{j=0}^{h-1} \psi_j^2.$$

(Operationally: *write the future and erase the future innovations.*)

(b) Here $\phi = \frac{1}{2}$ and $\theta = \frac{3}{10}$ (with $\Phi(B) = 1 - \frac{1}{2}B$, $\Theta(B) = 1 + \frac{3}{10}B$). The ψ -weights come from $\Theta(B)/\Phi(B) = \sum_j \psi_j B^j$:

$$\frac{1 + \frac{3}{10}B}{1 - \frac{1}{2}B} = \left(1 + \frac{3}{10}B\right) \sum_{k=0}^{\infty} \left(\frac{1}{2}\right)^k B^k.$$

Matching powers of B ,

$$\psi_0 = 1, \quad \psi_1 = \phi + \theta = \frac{1}{2} + \frac{3}{10} = \frac{4}{5}, \quad \psi_2 = \phi \psi_1 = \frac{1}{2} \cdot \frac{4}{5} = \frac{2}{5}$$

(in general $\psi_j = (\phi + \theta)\phi^{j-1}$ for $j \geq 1$). The point forecasts are obtained by zeroing future innovations in $X_{n+h} = \frac{1}{2}X_{n+h-1} + \varepsilon_{n+h} + \frac{3}{10}\varepsilon_{n+h-1}$:

$$\begin{aligned} X_{n+1}^n &= \frac{1}{2}X_n + \underbrace{\varepsilon_{n+1}}_{\rightarrow 0} + \frac{3}{10}\hat{\varepsilon}_n = \frac{1}{2}X_n + \frac{3}{10}\hat{\varepsilon}_n, \\ X_{n+2}^n &= \frac{1}{2}X_{n+1}^n + \underbrace{\varepsilon_{n+2}}_{\rightarrow 0} + \frac{3}{10}\underbrace{\varepsilon_{n+1}}_{\rightarrow 0} = \frac{1}{2}X_{n+1}^n = \frac{1}{4}X_n + \frac{3}{20}\hat{\varepsilon}_n. \end{aligned}$$

(For $h \geq 2$ the MA term has vanished and the forecast obeys $X_{n+h}^n = \frac{1}{2}X_{n+h-1}^n$, decaying geometrically to the mean 0.)

(c) With $\sigma^2 = 2$ and $\psi_0 = 1, \psi_1 = \frac{4}{5}, \psi_2 = \frac{2}{5}$, the forecast-error variances $P_{n+h}^n = \sigma^2 \sum_{j=0}^{h-1} \psi_j^2$ are

$$\begin{aligned} P_{n+1}^n &= \sigma^2 \psi_0^2 = 2 \cdot 1 = 2, \\ P_{n+2}^n &= \sigma^2 (\psi_0^2 + \psi_1^2) = 2 \left(1 + \frac{16}{25}\right) = 2 \cdot \frac{41}{25} = \frac{82}{25} = 3.28, \\ P_{n+3}^n &= \sigma^2 (\psi_0^2 + \psi_1^2 + \psi_2^2) = 2 \left(1 + \frac{16}{25} + \frac{4}{25}\right) = 2 \cdot \frac{45}{25} = \frac{18}{5} = 3.6. \end{aligned}$$

These increase toward the marginal variance

$$\gamma_0 = \sigma^2 \frac{1 + 2\phi\theta + \theta^2}{1 - \phi^2} = 2 \cdot \frac{1 + 2 \cdot \frac{1}{2} \cdot \frac{3}{10} + \frac{9}{100}}{1 - \frac{1}{4}} = 2 \cdot \frac{1 + \frac{3}{10} + \frac{9}{100}}{\frac{3}{4}} = 2 \cdot \frac{139/100}{3/4} = \frac{278}{75} \approx 3.707,$$

which is exactly $\sigma^2 \sum_{j=0}^{\infty} \psi_j^2$: as $h \rightarrow \infty$ the forecast loses all information and the error variance saturates at the unconditional variance γ_0 ($2 < 3.28 < 3.6 < 3.707$).

Problem 4 (5 points)

(a) Build $M(z) = I_2 - \Phi_1 z - \Phi_2 z^2$:

$$M(z) = \begin{pmatrix} 1 - 0.4z & 0 \\ -0.5z - 0.3z^2 & 1 - 0.9z + 0.2z^2 \end{pmatrix}.$$

This matrix is lower triangular, so its determinant is the product of the diagonal entries (the off-diagonal terms $\Phi_1^{(2,1)}, \Phi_2^{(2,1)}$ do *not* enter):

$$\det M(z) = (1 - 0.4z)(1 - 0.9z + 0.2z^2).$$

Roots. The first factor gives $z = 1/0.4 = 2.5$. The quadratic $0.2z^2 - 0.9z + 1 = 0$ has

$$z = \frac{0.9 \pm \sqrt{0.81 - 0.8}}{0.4} = \frac{0.9 \pm 0.1}{0.4} \implies z = 2.5 \text{ or } z = 2.$$

So the three roots are $z \in \{2.5, 2.5, 2\}$, all with $|z| > 1$ (strictly outside the unit circle). Hence the VAR(2) is **stable** (and therefore stationary). *Sanity check:* $\det M(0) = \det I_2 = 1$, as required.

Companion form. Stacking $Y_t = (X_t^\top, X_{t-1}^\top)^\top \in \mathbb{R}^4$ turns the VAR(2) into the VAR(1) $Y_t = FY_{t-1} + U_t$ with

$$F = \begin{pmatrix} \Phi_1 & \Phi_2 \\ I_2 & 0 \end{pmatrix} = \begin{pmatrix} 0.4 & 0 & 0 & 0 \\ 0.5 & 0.9 & 0.3 & -0.2 \\ 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \end{pmatrix}.$$

The companion test is “all eigenvalues of F strictly inside the unit circle”. It is the reciprocal of the root test, because $\det(I_2 - \Phi_1 z - \Phi_2 z^2) = z^4 \det(z^{-1}I_4 - F)$ up to a constant, so the eigenvalues of F are the reciprocals $1/z_i$ of the roots. Here $\lambda = 1/2.5 = 0.4$, $1/2.5 = 0.4$, $1/2 = 0.5$ (and a structural $\lambda = 0$ from the rank-deficient Φ_2), all with $|\lambda| < 1$: *same* conclusion, stable.

(b) Since $\Phi = \text{diag}(\frac{1}{2}, \frac{1}{4})$ has spectral radius $\frac{1}{2} < 1$, the VAR(1) is stable and back-substitution gives the matrix geometric (MA(∞)) filter

$$X_t = \sum_{j=0}^{\infty} \Phi^j \varepsilon_{t-j}, \quad \Phi^j = \text{diag}(2^{-j}, 4^{-j}).$$

By the VAR(1) covariance formula $\Gamma_0 = \sum_{k \geq 0} \Phi^k \Sigma (\Phi^k)^\top$ with $\Sigma = I_2$ (everything diagonal, so the channels decouple into independent AR(1)'s),

$$\Gamma_0 = \sum_{k=0}^{\infty} \text{diag}(4^{-k}, 16^{-k}) = \text{diag}\left(\frac{1}{1 - \frac{1}{4}}, \frac{1}{1 - \frac{1}{16}}\right) = \boxed{\text{diag}\left(\frac{4}{3}, \frac{16}{15}\right)}.$$

(Equivalently, the Lyapunov equation $\Gamma_0 = \Phi \Gamma_0 \Phi^\top + I_2$ gives componentwise $g_{jj} = \phi_j^2 g_{jj} + 1 \implies g_{jj} = 1/(1 - \phi_j^2)$, i.e. $1/(1 - \frac{1}{4}) = \frac{4}{3}$ and $1/(1 - \frac{1}{16}) = \frac{16}{15}$.) The lag-1 matrix follows from Yule–Walker $\Gamma_1 = \Phi \Gamma_0$:

$$\Gamma_1 = \text{diag}\left(\frac{1}{2}, \frac{1}{4}\right) \text{diag}\left(\frac{4}{3}, \frac{16}{15}\right) = \boxed{\text{diag}\left(\frac{2}{3}, \frac{4}{15}\right)}.$$

(c) Both Φ and $\Sigma = I_2$ are diagonal, so by the formula in (b) *every* Γ_τ is diagonal: all off-diagonal cross-covariances vanish at every lag. Hence the cross-spectrum $S_{1,2}(f) = \sum_\tau \gamma_\tau^{(1,2)} e^{-2\pi i f \tau} \equiv 0$, and the coherence

$$r_{1,2}(f) = \frac{S_{1,2}(f)}{\sqrt{S_{1,1}(f)S_{2,2}(f)}} = 0 \text{ for all } f.$$

The two channels are completely uncoupled: the off-diagonal entries of Φ (here zero) are exactly the cross-channel dynamics, and there is no shared innovation (Σ diagonal), so nothing links the series at any frequency.